

HW01



Final Grade

100 / 100

SUBMITTED 8/27/26, 1:39 PM

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100 / 100

1 MULTIPLE CHOICE CORRECT

10 / 10



For an airconditioning system, the thermostat is set at 68 degrees F and the current room temperature is 72 degrees F. What is the value of the error signal?

☐ A. 72☐ B. 4☒ C. **Correct:**
-4

Correct answer

☐ D. 68**2** MULTIPLE CHOICE CORRECT

10 / 10



The term is used for the element of a correction unit that provides the power to carry out the control action.

☒ A. **Correct:**
actuator

Correct answer

☐ B. sensor☐ C. process☐ D. reference

3

MULTIPLE CHOICE

CORRECT

10 / 10

Which of the following are elements of a mechatronic system?

- ☐ A. Actuators
- ☐ B. Sensors
- ☐ C. Microcontrollers\microprocessors
- ☒ D. **Correct:**
All the above

Correct answer



4

MULTIPLE CHOICE

CORRECT

10 / 10

Which of the following is not a typical attribute of open-loop control systems?

- ☒ A. **Correct:**
Relatively accurate in matching actual to required values
- ☐ B. Relatively simple
- ☐ C. Low cost
- ☐ D. Generally good reliability

Correct answer



5

MULTIPLE CHOICE

CORRECT

10 / 10

Microelectromechanical systems (MEMS) are mechanical devices built on semiconductor chips and usually consist of:

- ☐ A. Microprocessor

- ☐ B. Microsensors
- ☐ C. Microactuators
- ☒ D. **Correct:**
All the above

Correct answer

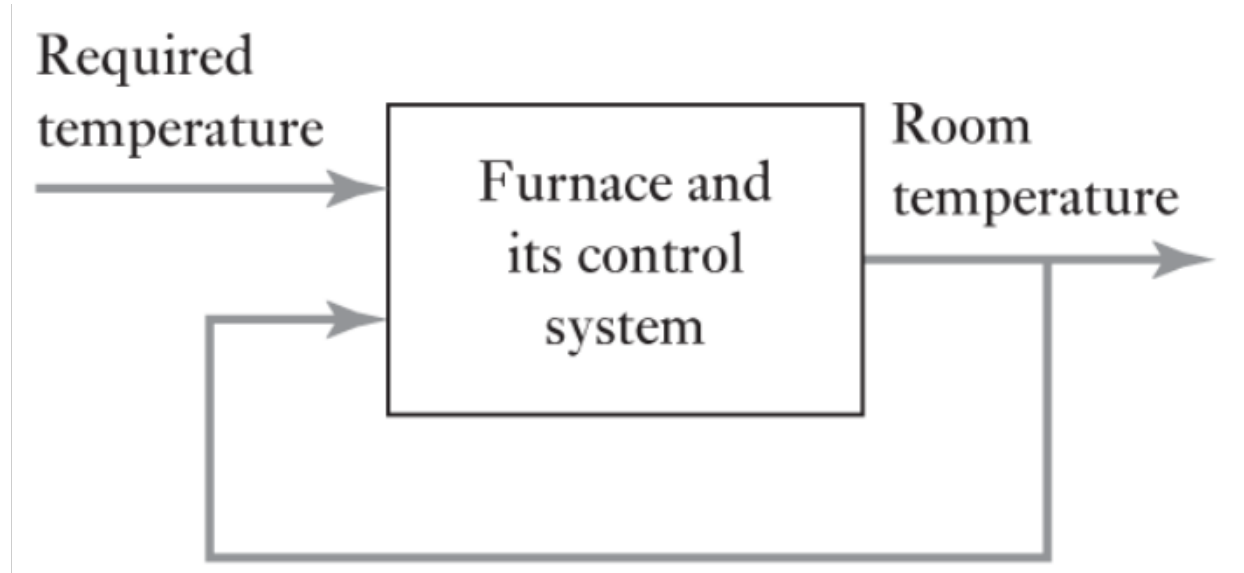
6

MULTIPLE CHOICE

CORRECT

10 / 10

The following picture is best defined as a:



- ☐ A. linear system
- ☒ B. **Correct:**
system with feedback control
- ☐ C. system with sequential control
- ☐ D. system with open loop control

Correct answer

7

MULTIPLE CHOICE

CORRECT

10 / 10

Analog signals are signals that are:

- ☐ A. between 0 and 10 volts
- ☐ B. boolean operators (on/off)
- ☐ C. sampled signal
- ☒ D. **Correct:**
continuous functions of time

Correct answer

8

MULTIPLE CHOICE

CORRECT

10 / 10



Embedded systems are:

- ☐ A. systems that talk to the internet (internet of things)
- ☒ B. **Correct:**
machines with micro controllers embedded into them
- ☐ C. purely mechanical systems
- ☐ D. open loop only

Correct answer

9

MULTIPLE CHOICE

CORRECT

10 / 10

A DAC is the device that converts

- ☒ A. **Correct:**
a digital signal into an analog signal.
- ☐ B. an analog signal into a digital signal.
- ☐ C. an analog signal to an error signal.

Correct answer

- ☐ D. a 12V signal to 5V signal.

10**TRUE/FALSE****CORRECT****10 / 10**

block diagrams represent the behavior of a real system by mathematical equations and are often described by a differential equation.

☒ True

Correct answer

☐ False



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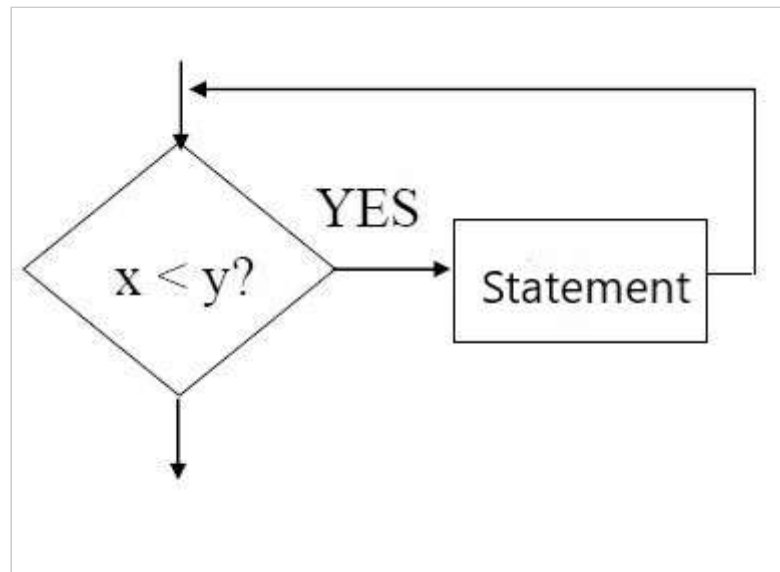
100 / 100

1

MULTIPLE CHOICE

CORRECT

10 / 10



For the repetition structure shown in the figure, which one of the following statements would not create an infinite loop?

☐ A. Display y

☒ B. **Correct:**
y=y-1

Correct answer

☐ C. $x=x-1$

☐ D. $y=y+1$

2

MULTIPLE CHOICE CORRECT

10 / 10



When designing a solution algorithm, a programmer should introduce unique names, which are

☐ A. Meaningful

☐ B. Use capital letter as separator (if the variable name is more than one word)

☐ C. Represent the variables or objects in the problem

☒ D. **Correct:** All the above

Correct answer

3

MULTIPLE CHOICE CORRECT

10 / 10



The process of identifying, isolating and resolving errors in software programs is called

☐ A. Coding

☐ B. Restructuring

☒ C. **Correct:** Debugging

Correct answer

☐ D. Fixing

4

FILL IN THE BLANK

CORRECT

10 / 10



In the following pseudocode, the user provides as input n=5. The output will be sum= **✓ 10**

Prompt 'Please input a number'

Get n

Initialize i to 1, sum to 0

While i < n do

sum = sum + i

i = i+1

end while

Display sum

Correct

Acceptable answer for **BLANK-1** is **10**

Responses must match exactly

5

TRUE/FALSE

CORRECT

10 / 10



In programming, logic errors prevent the code from being compiled or executed.

☐ True

☒ False

Correct answer

6 MULTIPLE CHOICE CORRECT

10 / 10



In a flow chart, Input/Output Operations:

- ☐ A. Are used to show either Sequence, Decision, Repetition, or Case.
- ☐ B. Are represented by rectangles; indicates a process such as a mathematical computation or variable assignment.
- ☒ C. **Correct:**
Are represented by parallelograms; indicate an input or output operation.
- ☐ D. Are represented by rounded rectangles; indicate a starting or ending point.

Correct answer

7 MULTIPLE CHOICE CORRECT

10 / 10



In a flow chart, a terminal:

- ☐ A. Is used to show either Sequence, Decision, Repetition, or Case.
- ☐ B. Is represented by rectangles; indicates a process such as a mathematical computation or variable assignment.

- ☐ C. Is represented by parallelograms; indicate an input or output operation.
- ☒ D. **Correct:**
Is represented by rounded rectangles; indicate a starting or ending point.

Correct answer

8

MULTIPLE CHOICE **CORRECT**

10 / 10



Pseudocode is based on:

- ☐ A. Arduino code (but scaled down for simplicity)
- ☐ B. Python
- ☒ C. **Correct:**
No coding language in particular as it is just instructions written in plain English.
- ☐ D. C (The programming language)

Correct answer

9

MULTIPLE CHOICE **CORRECT**

10 / 10



You develop a program that calculates taxes. What would be a proper variable name to express the adjusted gross income?

- ☒ A. **Correct:**
`adjustedGrossIncome`

Correct answer

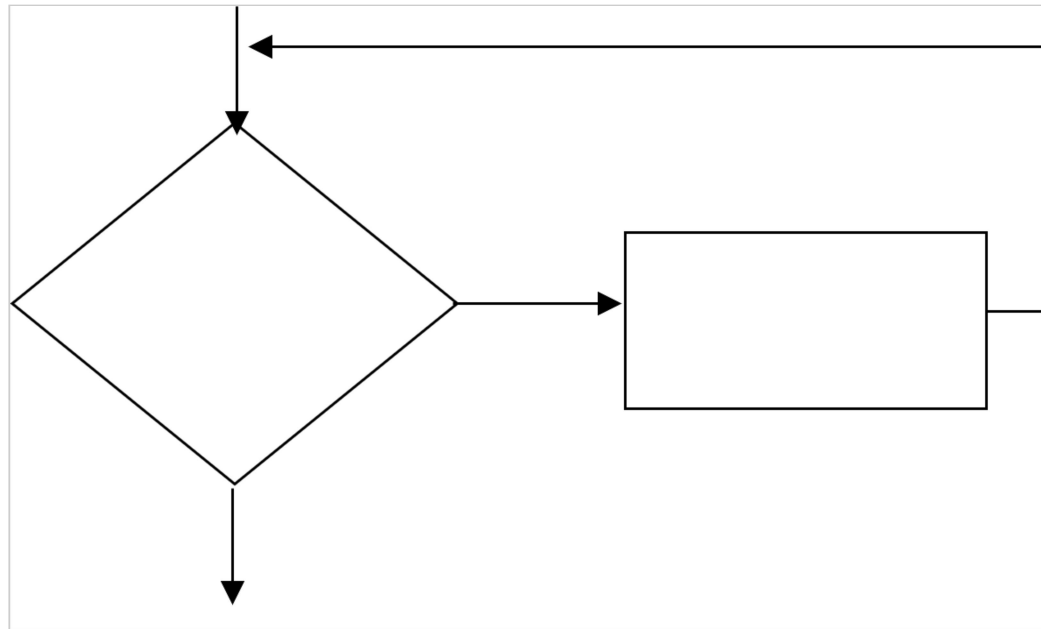
- ☐ B. x
- ☐ C. adjusted income
- ☐ D. int

10 **MULTIPLE CHOICE** **CORRECT**

10 / 10



What type of structure is this?



- ☐ A. Sequence
- ☐ B. Decision

☒ C. **Correct:** Repetition

Correct answer

☐ D. Case

HW03



Final Grade

80 / 100

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80 / 100

1

MULTIPLE CHOICE

CORRECT

10 / 10



Transducers can give different outputs for the same measured value depending whether this value is reached by continuously increasing or decreasing change. What kind of error is this?

- ☐ A. Non-linearity error
- ☐ B. Repeatability error
- ☒ C. **Correct:**
Hysteresis error
- ☐ D. Stability error

Correct answer

2

MULTIPLE CHOICE

CORRECT

10 / 10



Which of the following terms is not used to define the performance of a transducer?

- ☐ A. Sensitivity
- ☐ B. Stability
- ☐ C. Resolution
- ☒ D. **Correct:**
Size

Correct answer

3

MULTIPLE CHOICE

CORRECT

10 / 10

A potentiometer consists of a resistance element with a sliding contact to:

- ☐ A. Change the capacitance of the sensor.
- ☒ B. **Correct:**
Change the resistance of sensor.
- ☐ C. Change the current of the sensor.
- ☐ D. Change the position of the transducer.

Correct answer



4

TRUE/FALSE

CORRECT

10 / 10

The dynamic characteristics of a transducer refer to values given when steady-state conditions occur.

- ☐ True
- ☒ False

Correct answer



5

MULTIPLE CHOICE

CORRECT

10 / 10

A pressure sensor has a range of 0 to 1,000 kPa and a non-linearity error of $\pm 0.15\%$ of the full range. The error for a reading of 700 kPa is:

- ☐ A. ± 0.3 kPa
- ☐ B. ± 0.6 kPa
- ☐ C. ± 1.0 kPa

- ☒ D. **Correct:**
 ± 1.5 kPa

Correct answer

6

MULTIPLE CHOICE

CORRECT

10 / 10

Micro instruments make a strain gauge (MMF404146) with a resistance of 350 ohms and a gauge factor of 2.09. If the strain gauge, when applied to a surface, reads 350 Ohms before bending and 351 Oms after bending, what is the level of strain at the sensor? Note that the units of strain are change in length / original length and as such are very small.

- ☐ A. 1
- ☒ B. **Correct:**
0.00136
- ☐ C. 0.052
- ☐ D. 0.000050

Correct answer



7

CALCULATED NUMERIC

CORRECT

10 / 10

An incremental encoder has 180 slots on the disc. Express the resolution in degrees:

Your answer: 2

Correct

The answer is 2



8

MULTIPLE CHOICE

CORRECT

10 / 10

The range of a transducer is the:

- ☒ A. **Correct:**
Limit between which the input can vary.

Correct answer

- ☐ B. The maximum value of the input minus the minimum value.
- ☐ C. The difference between the result of the measurement and the true value.
- ☐ D. The extent to which the value may be wrong.

**9****CALCULATED NUMERIC****INCORRECT****0 / 10**

A shaft encoder is to be used with a 40mm radius tracking wheel to monitor linear displacement. If the encoder produces 512 pulses per revolution, what will be the number of pulses produced by a linear displacement of 150mm?

Your answer: 306

Incorrect

The answer is **305**

10**CALCULATED NUMERIC****INCORRECT****0 / 10**

The true speed of a car is 56 mph. The speedometer reading though is 54 mph. How much is the error in the measurement? Provide only the number without units.

HW04



Final Grade

90 / 100

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90 / 100

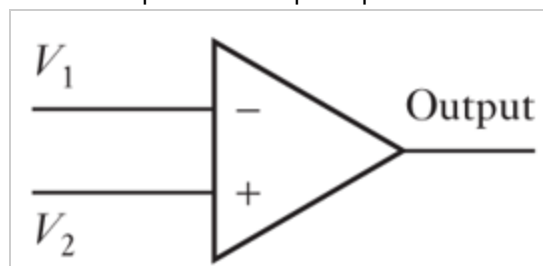
1 MULTIPLE CHOICE CORRECT

10 / 10



An op-amp is connected in a comparator configuration, as shown below. The positive power supply is $V_+ = 3.3\text{V}$ and the negative power supply is $V_- = -3.3\text{V}$.

If $V_1 = 2\text{V}$ and $V_2 = 0\text{V}$, what is the output of the op-amp?



☒ A. **Correct:**
-3.3V

Correct answer

☐ B. 3.3V

☐ C. 10V

☐ D. 0V

2 MULTIPLE CHOICE CORRECT

10 / 10



From the list below, select the statements that apply to ideal operational amplifiers:

☒ A. **Correct:**
They have infinite gain.

Correct answer 50%

☐ B. **Correct:**

- ☒ B. They have infinite input impedance. Correct answer 50%
- ☐ C. They have infinite output impedance. -50%
- ☐ D. They require offset voltage. -50%
- 

3 MULTIPLE CHOICE CORRECT

10 / 10

A thermocouple is used with a compensation circuit:

- ☐ A. Because the e.m.f. is not a linear function of temperature.
- ☐ B. To increase the time constant of the sensor.
- ☐ C. To amplify the small thermoelectric e.m.f.
- ☒ D. **Correct:**
Because the cold junction is not held at 0°C.

Correct answer



4 MULTIPLE CHOICE CORRECT

10 / 10

In general, all other things equal, you want a sensor with:

- ☐ A. High sensitivity, low linearity, and high hysteresis.
- ☐ B. Low sensitivity, low linearity, and high hysteresis.
- ☒ C. **Correct:**
High sensitivity, high linearity, and low hysteresis.
- ☐ D. Low sensitivity, low linearity, and low hysteresis .

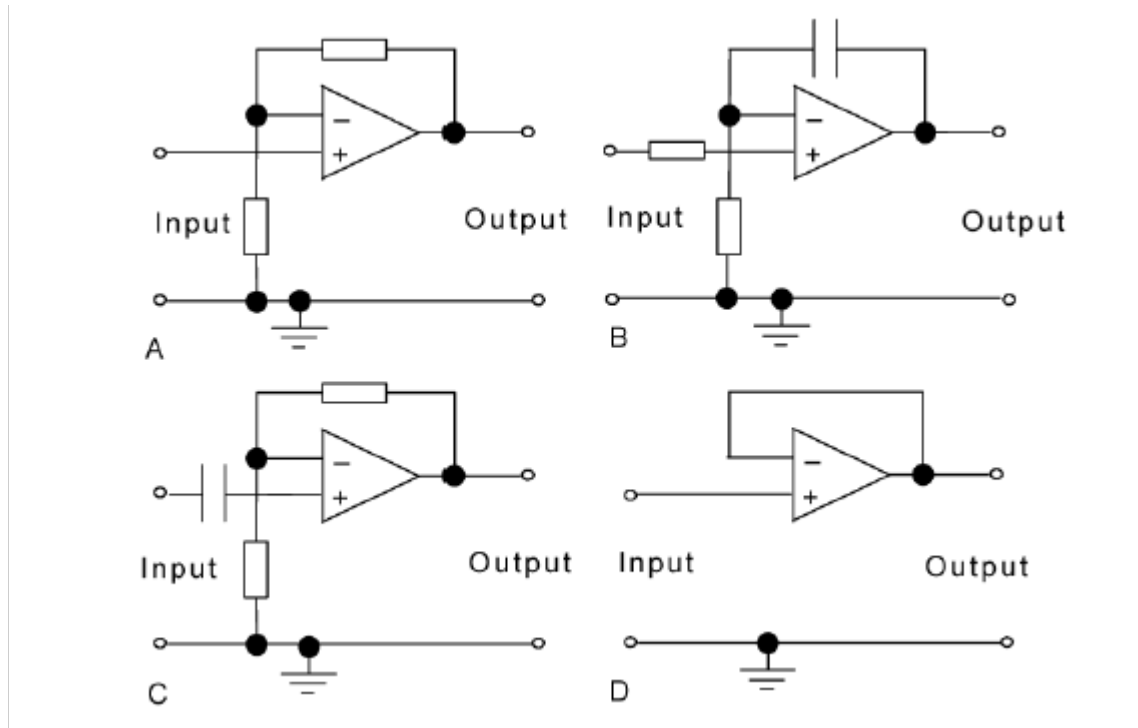
Correct answer



5 MULTIPLE CHOICE CORRECT

10 / 10

Select the operational amplifier circuit configuration that is a voltage follower. This means the output voltage directly follows the input voltage.



- ☐ A. Image A
- ☐ B. Image B
- ☐ C. Image C
- ☒ D. **Correct:**
Image D

Correct answer

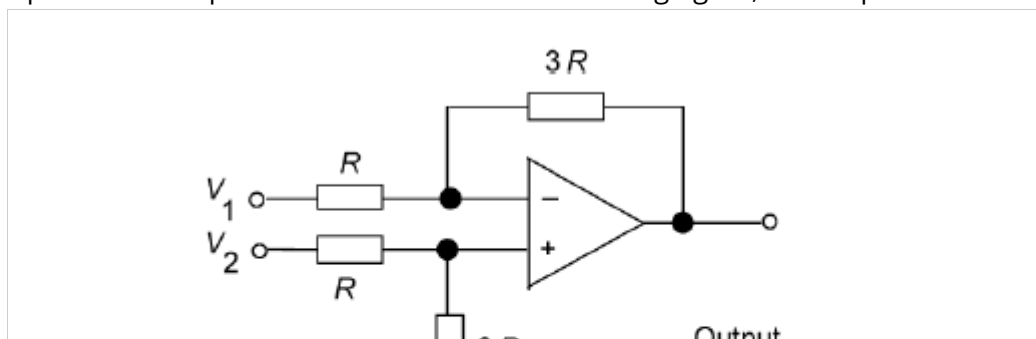
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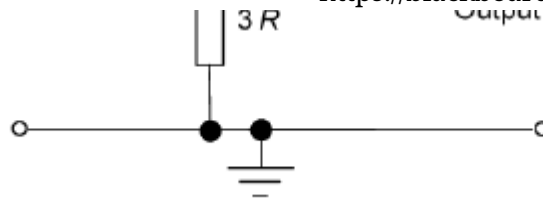
MULTIPLE CHOICE

CORRECT

10 / 10

For the operational amplifier circuit shown in the following figure, the output is:





- ☐ A. Three times the sum of V_1 and V_2 .
- ☒ B. **Correct:** Three times the difference between V_1 and V_2 .
- ☐ C. Twice the sum of V_1 and V_2 .
- ☐ D. Twice the difference between V_1 and V_2 .

Correct answer

7

MULTIPLE CHOICE

INCORRECT

0 / 10

We need to track rapid temperature changes in a very small space. What sensor should we use?

- ☐ A. Thermistor
- ☒ B. **Incorrect:** Thermocouple
- ☐ C. Bimetallic thermostat
- ☐ D. Photodiode

Correct answer

8

MULTIPLE CHOICE

CORRECT

10 / 10

A photoresistor

- ☐ A. uses two different metals joined together at a junction to generate a difference in

potential.

Correct:

- ☒ B. has a resistance which depends on the intensity of the light falling on it. Correct answer
- varies the voltage across the junction between the base and the emitter (in a
- ☐ C. transistor) as a function of temperature and therefore can be used as a measure of temperature.
- ☐ D. has a light-sensitive collector-base p-n junction and can be used to detect light.



9 MULTIPLE CHOICE CORRECT

10 / 10

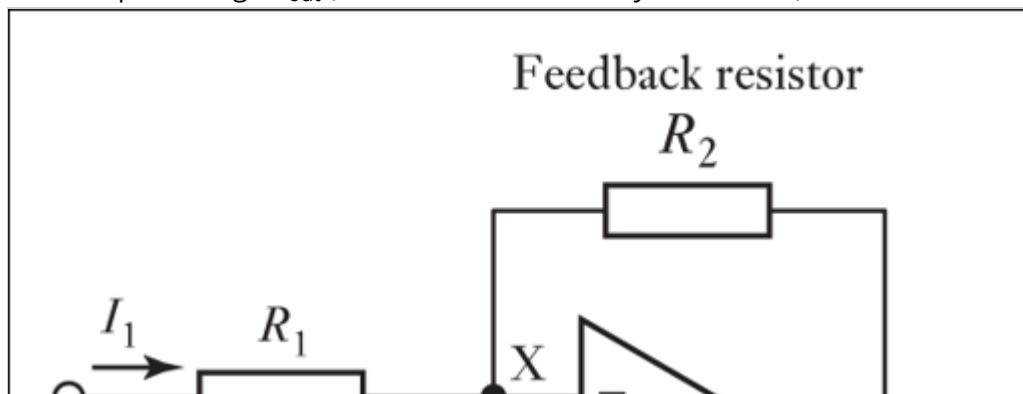
A thermocouple sensor has a time constant of 50 seconds in still air. When it is subject to a temperature change of 100°C, in 50 s its temperature will have changed by about:

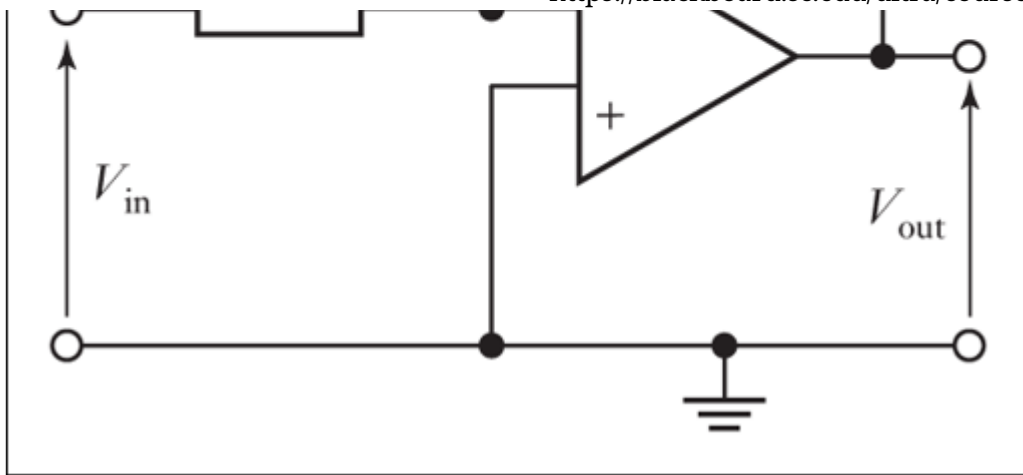
- ☐ A. 31°C
- ☐ B. 50°C
- ☒ C. **Correct:**
63°C Correct answer
- ☐ D. 100°C

10 CALCULATED NUMERIC CORRECT

10 / 10

Consider the operational amplifier in the configuration below. If $V_{in}=5V$, $R_1=10\Omega$ and $R_2=50\Omega$, calculate the output voltage V_{out} (do not include units in your answer).





HW05



Final Grade

100 / 100

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100 / 100

1

MULTIPLE CHOICE

CORRECT

10 / 10



A sensor gives an analog output range of 12V. How many bits are needed for an ADC if there is to be a resolution of at least 20 mV?

☐ A. 8☐ B. 9☒ C. **Correct:**
10

Correct answer

☐ D. 12

2

MULTIPLE CHOICE

CORRECT

10 / 10



A sensor, for an input of 4 V, gives a digital output of a word of 8 bits. The resolution is:

☐ A. $4/8$ V☐ B. $8/4$ V☒ C. **Correct:**
 $4/2^8$ V

Correct answer

☐ D. $2^8/4$ V

3

MULTIPLE CHOICE

CORRECT

10 / 10

Which of the following binary words represents the decimal number 189?

- ☐ A. 11011011
- ☒ B. **Correct:**
10111101
- ☐ C. 10111100
- ☐ D. None of the above.

Correct answer



4

MULTIPLE CHOICE

CORRECT

10 / 10

The 5-bit binary number "10010" is what in decimal?

- ☐ A. 5
- ☐ B. 9
- ☒ C. **Correct:**
18
- ☐ D. 24

Correct answer



5

MULTIPLE CHOICE

CORRECT

10 / 10

You need to filter a signal and block all frequencies from 100Hz to 200Hz. What kind of filter will you use?

- ☐ A. low-pass filter

- ☐ B. high-pass filter
- ☐ C. band-pass filter
- ☒ D. **Correct:**
band-stop filter

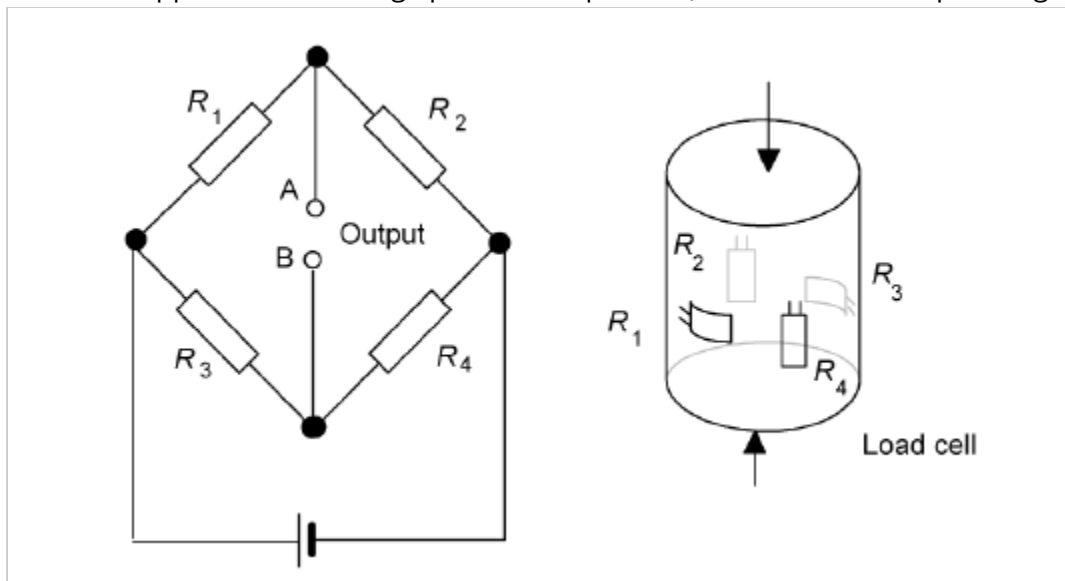
Correct answer

6

MULTIPLE CHOICE CORRECT

10 / 10

Consider the full bridge made of strain gauges attached to a metal cylinder as shown below. When a compressive load is applied to the metal cylinder the resistance of R_1 and R_3 will increase due to the stretching of the strain gauge while R_2 and R_4 will decrease due to the compression of the strain gauge. Given that the voltage supplied by the battery remains constant, what happens to the voltage potential at point A (measured with respect to ground)?



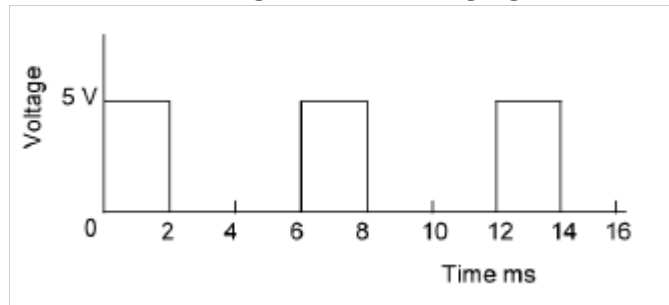
- ☐ A. Goes up
- ☒ B. **Correct:**
Goes down
- ☐ C. Does not change
- ☐ D. Goes sideways

Correct answer

7 MULTIPLE CHOICE CORRECT

10 / 10

What pulse width modulation is used to give the following signal from a 5V DC input?



- ☒ A. **Correct:**
33.3%
- ☐ B. 83.3%
- ☐ C. 166.7%
- ☐ D. 500%

Correct answer

8 MULTIPLE CHOICE CORRECT

10 / 10

Which of the following statements about filtering is not true?

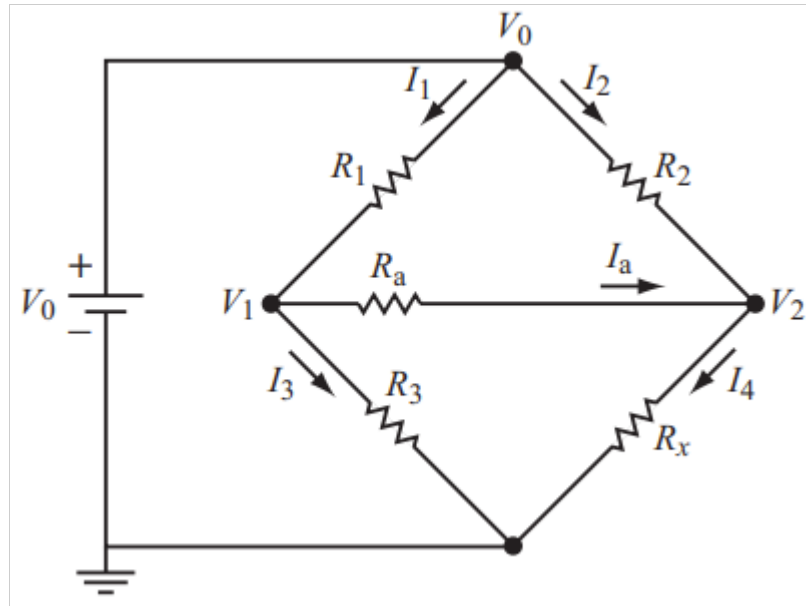
- ☐ A. Passive filters use only resistors, inductors and capacitors.
- ☐ B. The range of frequencies passed by a filter is known as the pass band.
- ☒ C. **Correct:**
Operational amplifiers are used in filters to amplify the signal.
- ☐ D. The cut-off frequency is defined as being that at which the output voltage is 70.7% of that in the pass band.

Correct answer

9 MULTIPLE CHOICE CORRECT

10 / 10

For the Wheatstone bridge circuit shown below, if $R_1 = 1\ \Omega$, $R_2 = 2\ \Omega$, and $R_x = 3\ \Omega$, to what value should R_3 be adjusted so as to achieve a balanced condition?



☒ A. **Correct:**
1.5 Ω

Correct answer

☐ B. 2.5 Ω

☐ C. 3.5 Ω

☐ D. 4.5 Ω

10 **MULTIPLE CHOICE** **CORRECT**

10 / 10

The decimal system is a:

☐ A. base-1 system

☐ B. base-2 system

Correct:

HW06



Final Grade

100 / 100

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100 / 100

1 MULTIPLE CHOICE

10 / 10



What is the purpose of an anti-aliasing filter placed before an ADC?

- ☐ A. Detect transmission errors in digital data.
- ☐ B. Convert analog signals to digital signals.
- ☐ C. Increase the sampling frequency.
- ☒ D. Block high-frequency components that could cause aliasing.

2 MULTIPLE CHOICE

10 / 10



The acronym DAQ stands for:

- ☐ A. Data availability qualifier.
- ☒ B. Data acquisition.
- ☐ C. Device aquatic.
- ☐ D. Digital analog converter.

3 MULTIPLE CHOICE

10 / 10



The "quantization error" in a ADC comes from:

- ☐ A. The quality of the ADC.
- ☒ B. The step like nature of digitizing the analog signal.
- ☐ C. It is not related to the ADC at all.
- ☐ D. The aliasing of the signal.



4 MULTIPLE CHOICE

10 / 10

Which of the following statements is not true about the Nyquist frequency?

- ☐ A. It guarantees that the reconstructed signal is free of aliasing.
- ☒ B. It rebuilds the signal capturing both the frequency and amplitude components.
- ☐ C. It is the theoretical minimum number of sample points required.
- ☐ D. It is named after Harry Nyquist.



5 MULTIPLE CHOICE

10 / 10

In an even-parity system, what parity bit should be added to the data word **1011**?

- ☐ A. 0
- ☐ B. 10
- ☐ C. No parity bit is needed

☒ D. 1

6 MULTIPLE CHOICE

10 / 10

Why is a sample-and-hold circuit used before or within an ADC system?

- ☐ A. To convert digital output into analog output.
- ☐ B. To eliminate the need for a clock signal.
- ☒ C. To hold the input voltage constant during conversion.
- ☐ D. To increase the number of ADC bits.



7 MULTIPLE CHOICE

10 / 10

A 10-bit ADC has an input range of 0–10 V. What is its approximate resolution?

- ☒ A. 9.8 mV
- ☐ B. 1.0 V
- ☐ C. 0.98 mV
- ☐ D. 98 mV



8 MULTIPLE CHOICE

10 / 10

To rebuild a sinusoidal signal, accurate in both frequency and amplitude, what is the minimum sampling required?

- ☒ A. The Nyquist frequency + 1, or $\frac{1}{2}$ of the sampling frequency $f_s + 1$ sample
- ☐ B. The Nyquist frequency, or $\frac{1}{2}$ of the sampling frequency f_s
- ☐ C. 2 times the sampling frequency f_s
- ☐ D. 10 times the sampling frequency f_s

9 MULTIPLE CHOICE

10 / 10



One advantage of digital signals over analog signals is that digital signals are generally:

- ☐ A. More sensitive to small voltage fluctuations.
- ☐ B. Continuous in time.
- ☐ C. Unable to be processed by microprocessors.
- ☒ D. Less affected by noise, drift, and component tolerances.

10 MULTIPLE CHOICE

10 / 10

A multiplexer:

- ☐ A. Is used for collecting the parity bits for validating the transfer of information.
- ☐ B. Is a system that converts an analog signal, such as a sound picked up by a microphone or light entering a digital camera, into a digital signal.
- ☒ C. Is a circuit that is able to have inputs of data from a number of sources and then, by selecting an input channel, give an output from just one of them.

☐ D. Is a system used for taking data from sensors and inputting that data into a



- D.** computer for processing.